1. Identity

Document	STAR-PKG-STAR-ANOMALY-EN · v1.5 · 2026-10-20
Package	star-anomaly
Version	0.1.0
Capability	star.orbit.anomalies
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_anomaly

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_anomaly-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_anomaly-0.1.0-py3-none-any.whl	8aba738e1fe4b0aefe0b6841288ea2fb06c28829341675b4218e4c5f0e982620
star_anomaly.py	76f3de92f09322b24e3c45c652b3063583486a03b023535c152ab6489a7f44ea
star_anomaly-0.1.0-source.zip	691261d5657a59524c9aa4109e27773fbad54747e8ab0e4a3227e53bd38a60a6

5. What it does, and its limits

Kepler's equation and the conversions between mean, eccentric, hyperbolic and true anomaly for closed and open orbits: a bracketed Newton solver that cannot diverge, a residual that does not cancel near the periapsis of nearly parabolic orbits, the revolution kept; within 1e-14 rad of 50-digit arithmetic on 1120 orbits.

The parabola (e exactly 1) is not covered: use Barker's equation elsewhere. Some conversions are badly conditioned by nature and no code can repair that: near a parabola, the eccentric anomaly recovered from a true anomaly close to 180 degrees, and the hyperbolic anomaly recovered from a true anomaly close to the asymptote, hold fewer digits than the input (the function is accurate for the float it is given; the float does not hold the information). The 50-digit reference uses other formulas than the module's but was written by the same author. No propagation in time: converting a time to a mean anomaly is one multiplication, left to the caller.

- Closed orbits (0 <= e < 1): `eccentric_from_mean`, `mean_from_eccentric`, `true_from_eccentric`, `eccentric_from_true`, `true_from_mean`, `mean_from_true`. Open orbits (e > 1): `hyperbolic_from_mean`, `mean_from_hyperbolic`, `true_from_hyperbolic`, `hyperbolic_from_true`. Angles in radians.
- The closed-orbit functions keep the revolution (nothing is wrapped). Kepler's equation is solved by Newton's method inside a bracket that always contains the root, with the residual written so that it does not cancel near the periapsis of a nearly parabolic orbit; the true anomaly comes from a form with no quadrant to choose.
- Measured on 1120 orbits (620 closed, 500 open, a fifth of them nearly parabolic) and two published examples, four conversions each: within 1e-14 rad of 50-digit arithmetic on every case; within 2e-14 rad of hapsira and Orekit on the ordinary orbits (the nearly parabolic ones are reported apart in the evidence file).

- Every function returns a float or raises `ValueError`: values that are not finite real numbers, an eccentricity on the wrong side of 1 for the function called, an angle beyond 1e9 rad, a true anomaly at or beyond the asymptote of an open orbit, a result too large for a float.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_anomaly_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-anomaly}: Mean, eccentric, hyperbolic and true anomaly},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_anomaly/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts